Ethics, Privacy, and Security in Data Science and AI (CST4235)

Week 5

(15 Credits)

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Bias and Fairness

Today's Agenda

- **What is fairness?**
- **How to identify some biases for fairness?**
  - Missing Feature Values
  - Unexpected Feature Values
  - Data Skew
- **Metrics to measure fairness**
  - Confusion matrix
  - Equal opportunity
  - Equalized odds



Fairness in AI

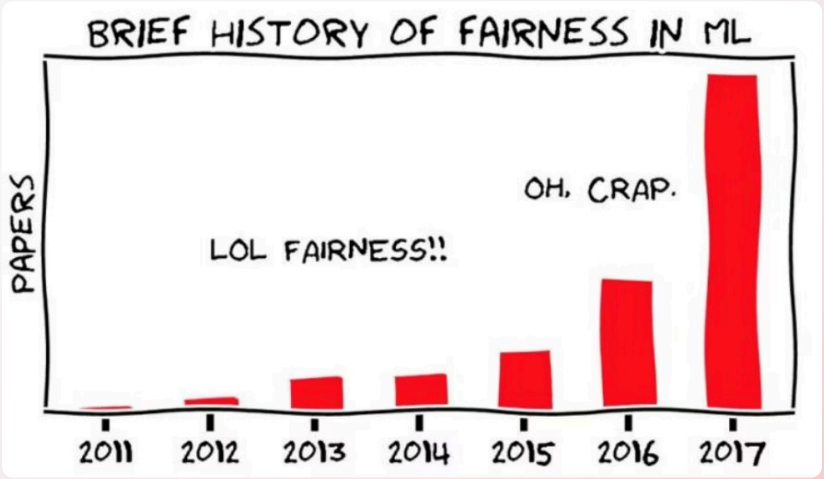
Algorithms learn from human data – and human data often carries bias.
When we encode that bias into systems, unfair patterns get automated at scale.

Bias in AI isn't always intentional; it can arise from:

- Skewed or incomplete datasets
- Poorly defined objectives
- Lack of representation or context awareness

A Brief History of Fairness in ML

Early ML Era - "LOL Fairness" → Fairness wasn't even discussed.
Applied ML Era - "Oh crap!" → Real-world bias exposed serious harm.
Modern Era - "We need fairness research" → A new field emerges to make AI accountable.



Defining Fairness

Goal: Create a measure that helps us check if a machine learning model makes fair decisions.

Key Terms

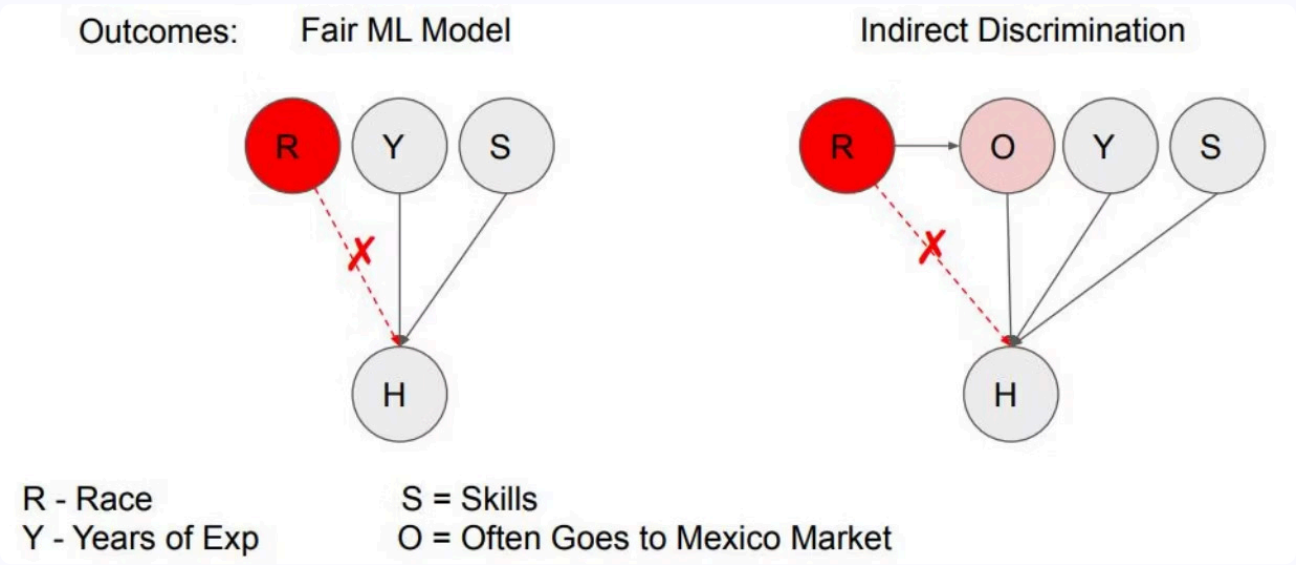
- **Y** → The *true* outcome (e.g., loan approved = 1, loan denied = 0)
- **C** → The *model's prediction*
- **A** → The *protected attribute* (e.g., gender, race, age)
  - **A = 0:** Privileged group *(e.g., male)*
  - **A = 1:** Unprivileged group *(e.g., female)*

We compare the model's predictions (**C**) with the protected attribute (**A**)
→ to see whether the model treats groups **differently** just because of who they are.

If predictions depend on A, bias exists.
If predictions are independent of A, the model behaves fairly.

Fairness through Unawareness

Even if we remove sensitive attributes **(like race or gender)** from our dataset, bias can still sneak in through other features that are correlated with them.



Fair ML Model (Left Diagram)

- **R** (Race) has no direct or indirect influence on the hiring decision (**H**).
- The model uses fair indicators like:
  - **Y** (Years of Experience)
  - **S** (Skills)
- So, no discrimination occurs because race is not affecting the outcome.

Indirect Discrimination (Right Diagram)

- Race (**R**) doesn't directly affect hiring (**H**) – but it influences another variable, like **O** (Often Goes to Mexico Market).
- The model uses **O** in decision-making, unintentionally reintroducing bias.
- This is called proxy discrimination – when a variable indirectly reflects a protected attribute.

Example

A company removes "race" from a hiring dataset.
But it includes "zip code" – which correlates strongly with racial demographics.
Even without race, the model still discriminates because zip code acts as a proxy.

Missing Feature Values

When building models, missing or incomplete data doesn't just hurt accuracy – it can also introduce unfairness.

	longitude	latitude	total_rooms	population	households	median_income	median_house_value
count	17000.0	17000.0	17000.0	3000.0	3000.0	3000.0	17000.0
mean	-119.6	35.6	2643.7	1429.6	501.2	3.9	207.3
std	2.0	2.1	2179.9	1147.9	384.5	1.9	116.0
min	-124.3	32.5	2.0	3.0	1.0	0.5	15.0

Example: California Housing Dataset

- Each column represents a feature (like longitude, rooms, or income).
- The yellow columns have full data (17,000 records).
- The red columns – *population, households, median_income* – have only 3,000 values.
- This means 14,000 rows are missing data in those features.

Problem:

- If missing values come mostly from certain areas or income groups,
  the model learns less about them – and its predictions become biased.
- For example, if median income data is missing from low-income regions,
  the model may overestimate house prices **(predicts a value that's higher than the real one)** there,
  favouring wealthier neighbourhoods.

Unexpected Feature Values.....What to Look For?

	longitude	latitude	total_rooms	population	households	median_income	median_house_value
1	-121.7	38.0	7105.0	3523.0	1088.0	5.0	0.2
2	-122.4	37.8	2479.0	1816.0	496.0	3.1	0.3
3	-122.0	37.0	2813.0	1337.0	477.0	3.7	0.3
4	-103.5	43.8	2212.0	803.0	144.0	5.3	0.2
5	-117.1	32.8	2963.0	1162.0	556.0	3.6	0.2

Example: California Housing Dataset

- Each row represents a location in California.
- But **row 4** has coordinates **(-103.5, 43.8)** – that's actually **South Dakota**, not California!

If your model accidentally includes that row, it may think:

"Oh, houses in this 'California' area are very cheap!"

But in reality, that's not California data at all – it's from another region with completely different housing prices, income levels, and population.

So when the model learns from this incorrect example, it may:

- Predict **lower house prices** for real California homes (biased prediction).
- Fail to learn accurate **regional patterns**.

Data Skew – When Data Doesn't Represent Reality

Example: California Housing Dataset

- The blue and red dots show median house prices, from *low (blue)* to *high (red)*.
- Notice how almost all data points are concentrated in Northern and Central California, while the southern half is empty.

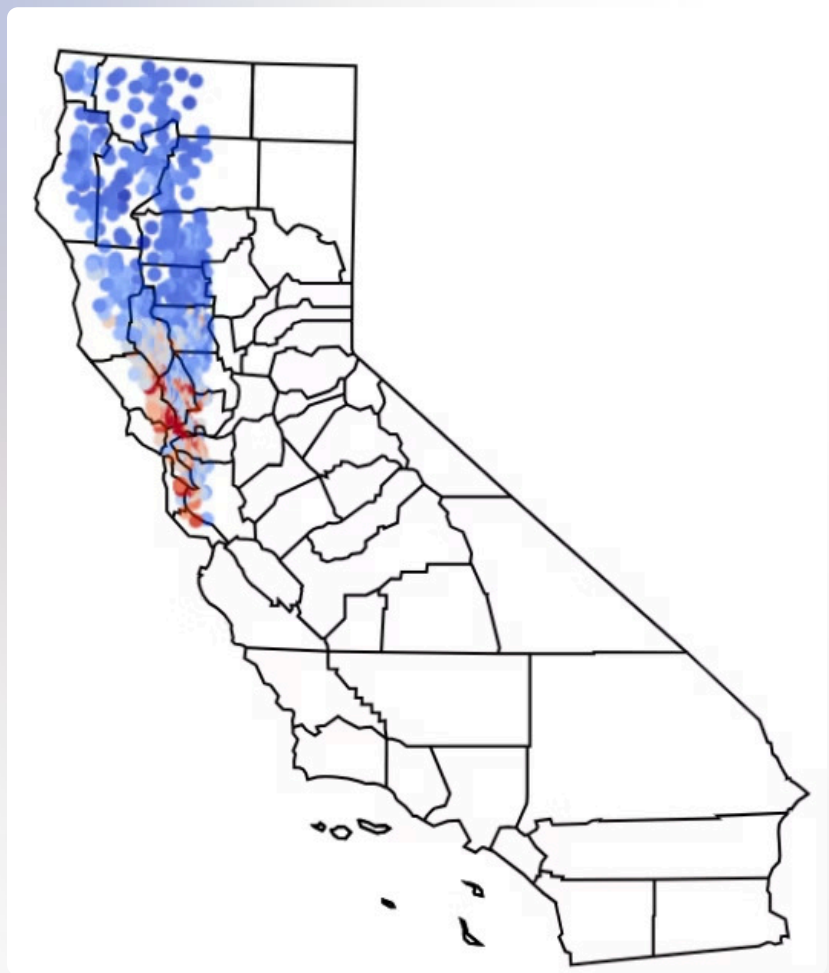
So, the model learns rules like:

"Most houses near the coast aren't too expensive,"
"Bigger houses in rural areas cost less,"
because that's true in the north — but not true in the south.

Problem: When you later ask the model to predict house prices in **Los Angeles** or **San Diego**, it still uses the **northern rules** it learned – because it doesn't know anything about the south.

That leads to:

- **Underestimation:** It predicts prices too low for the south (since homes there are costlier).
- **Overfitting:** It assumes northern trends (like smaller populations or lower income levels) apply everywhere.
- **Bias:** The model gives unfair or inaccurate results for people living in unrepresented areas (southern California).



Metrics to Measure Fairness

Fairness depends on the type of problem your model is solving:

- **Regression** - predicting a continuous value
  Example: Estimating house prices or salary levels.
  → Check if the model's errors are similar across different groups.
  (e.g., does it overpredict for men but underpredict for women?)
- **Classification** - predicting a category or label
  Example: Approving or rejecting a loan, hiring or not hiring a candidate.
  → Compare the rate of positive outcomes (like approvals) between groups.
- **Clustering** - grouping similar data points together
  Example: Grouping customers by spending behaviour.
  → Ensure groups are not formed along sensitive attributes (like race or gender).

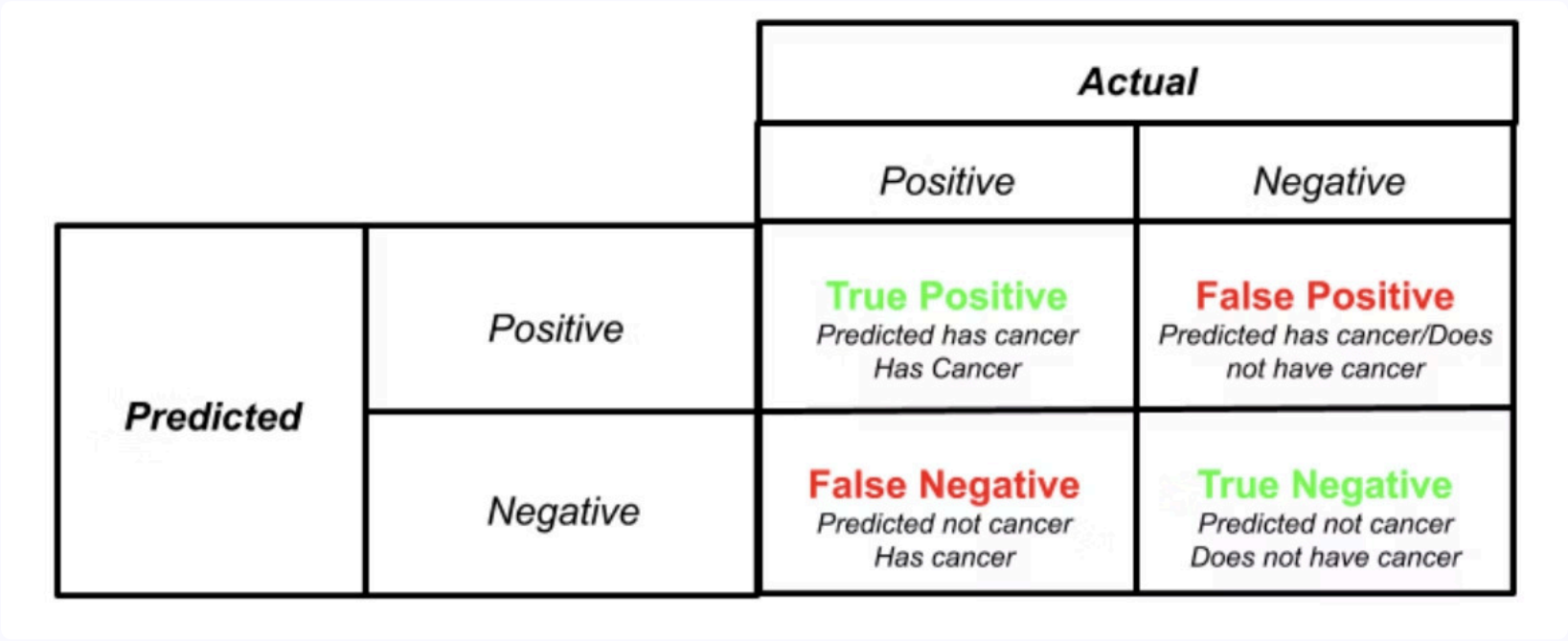
In binary decision tasks (e.g., loan approval):

- The model makes a **yes/no** decision – approve or deny.
- To test fairness, we compare how often each group receives a positive decision.
  - If 80% of male applicants are approved but only 50% of female applicants are,
    → the model is biased.

Fairness metrics help quantify and prove whether such differences exist.

Measuring Fairness in Classification - **Confusion Matrix**

- A confusion matrix is a table that helps us understand how well a classifier performs by showing:



  - **True Positives (TP):** Correctly predicted positive cases
  - **True Negatives (TN):** Correctly predicted negative cases
  - **False Positives (FP):** Incorrectly predicted as positive
  - **False Negatives (FN):** Incorrectly predicted as negative
- By comparing **TP, FP, TN, and FN** across different groups, we can see if the model performs **unequally**.

Confusion Matrix - Accuracy

$$\text{Accuracy} = \frac{TN + TP}{N}$$

$$\text{where } N = TN + FP + FN + TP$$

Scenario: A model is developed to predict the presence of tumours using a dataset of 1,000 patients – 500 female and 500 male medical records.

Model results:

- **True Positives (TP):** 16 – correctly predicted tumour cases
- **False Positives (FP):** 4 – predicted tumour, but actually healthy
- **False Negatives (FN):** 6 – missed tumour cases
- **True Negatives (TN):** 974 – correctly predicted healthy cases

Accuracy tells you the overall proportion of correct predictions made by the model.

Confusion Matrix - Precision

True Positives (TPs): 16	False Positives (FPs): 4
False Negatives (FNs): 6	True Negatives (TNs): 974

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{16}{16 + 4} = 0.800$$

Scenario: Continuing the same tumour prediction model with 1,000 patients:

Model results:

- **True Positives (TP):** 16 – correctly predicted tumour cases
- **False Positives (FP):** 4 – predicted tumour, but actually healthy
- **False Negatives (FN):** 6 – missed tumour cases
- **True Negatives (TN):** 974 – correctly predicted healthy cases

Precision:* *80% correct* *BUT* *Some people are incorrectly flagged as having the condition when they actually don't.

Confusion Matrix — Recall (Sensitivity)

True Positives (TPs): 16	False Positives (FPs): 4
False Negatives (FNs): 6	True Negatives (TNs): 974

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{16}{16 + 6} = 0.727$$

Scenario: Continuing the same tumour prediction model with 1,000 patients:

- **True Positives (TP):** 16 – correctly detected tumour cases
- **False Negatives (FN):** 6 – missed tumour cases
- **False Positives (FP):** 4 – predicted tumour but actually healthy
- **True Negatives (TN):** 974 – correctly predicted healthy

Recall: 72.7% correct BUT Some people with the condition are still being missed.

Confusion Matrix (Fairness Across Genders)

When we break results down by gender, the model performs very differently:

Female Patient Results

True Positives (TPs): 10	False Positives (FPs): 1
False Negatives (FNs): 1	True Negatives (TNs): 488

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{10}{10 + 1} = 0.909$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{10}{10 + 1} = 0.909$$

Male Patient Results

True Positives (TPs): 6	False Positives (FPs): 3
False Negatives (FNs): 5	True Negatives (TNs): 486

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{6}{6 + 3} = 0.667$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{6}{6 + 5} = 0.545$$

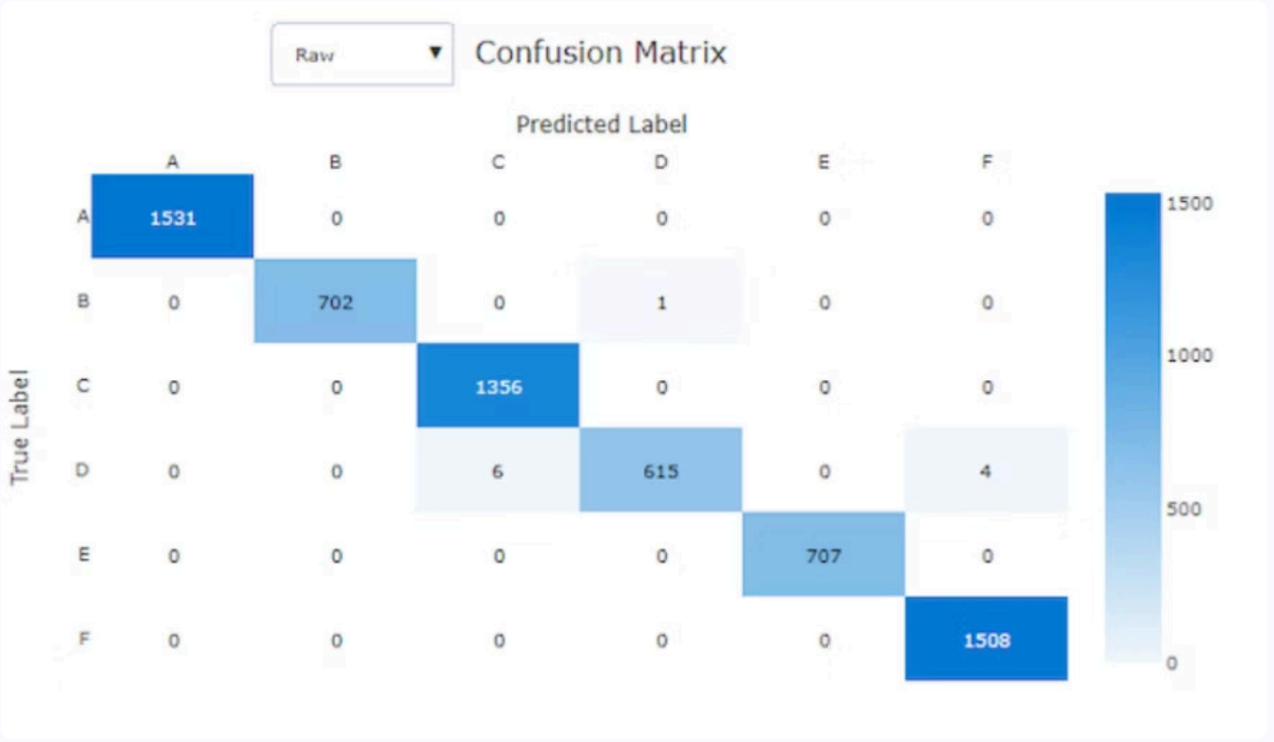
- For **female patients**:
  - **Recall: 90.9%** – the model correctly finds almost all true tumour cases, missing only a few.
  - **Precision: 90.9%** – when it predicts a tumour, it's right most of the time.
- For **male patients**:
  - **Recall: 54.5%** – the model finds only about half of the actual tumour cases, missing nearly the other half.
  - **Precision: 66.7%** – even among tumour predictions, one in three is wrong.

- This shows the model is less accurate for male patients, meaning it could miss or misclassify many male cases.
- In fairness terms, the model is biased toward female patients because it performs much better for them.

However

- We can argue that the model is not necessarily "unfair" on purpose – it simply reflects **data scarcity** for male cases.
- This is an excellent teaching example of **representation bias**: when one group dominates the data, the model underperforms for others.

Confusion Matrix

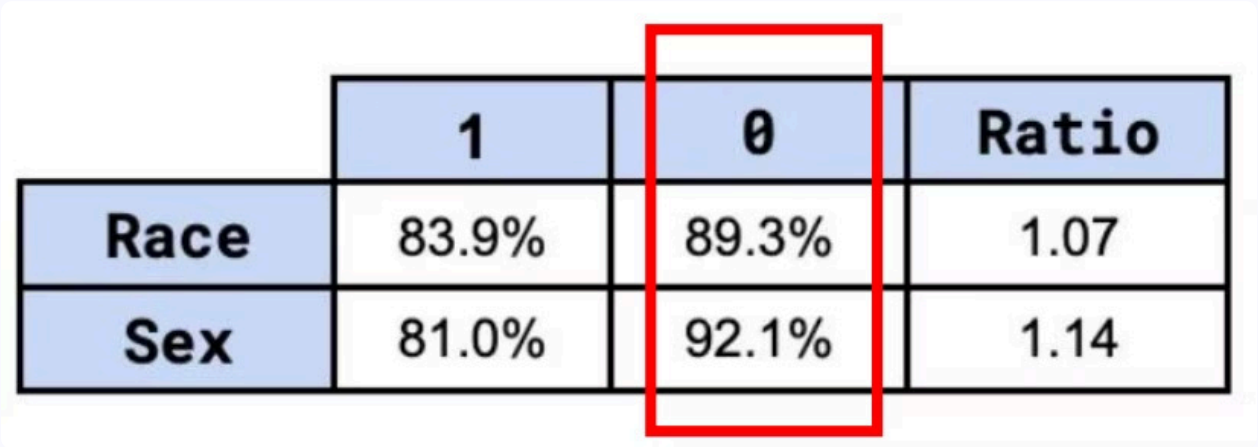


- Each row represents the **actual class**, and each column represents the **predicted class**.

For example, a value in row D and column F means the model predicted class F when it should've been D – that's a misclassification.

- Ideally, a perfect model would have all values only along the **diagonal** – meaning all true labels match predictions.
- In Figure, most values are indeed along the diagonal – bright blue boxes – which means the model performs very well overall.
- The name 'confusion matrix' comes from – it captures how and where a model gets *confused* between classes.

Measuring Fairness in Classification



	1	0	Ratio
Race	83.9%	89.3%	1.07
Sex	81.0%	92.1%	1.14

- A classifier predicts if a person's income is above or below $50K.
- 1 represents a privileged group, and 0 represents an unprivileged group – those who have historically faced disadvantage or bias.
- To assess fairness, we compare accuracy for each group using protected features such as race and sex.
- At first glance, the model seems to perform better for the unprivileged group (higher accuracy).
  However, accuracy alone can be misleading – it doesn't reveal *why* these differences occur.
  The model could still be biased in recall, precision, or false negative rates.
- A fair model should show consistent performance across groups, not just in accuracy but also in precision, recall, and error distribution.

Measuring Fairness in Classification

	1	0	Ratio
Race	83.9%	89.3%	1.07
Sex	81.0%	92.1%	1.14

- A False Positive occurs when the model predicts a person will repay a loan – but they actually won't.
- It reduces accuracy, yet the person still benefits by receiving the loan.
- This seems harmless, but it becomes unfair if certain groups (e.g., based on race or sex) receive more of these beneficial mistakes.
- Meanwhile, other groups may face False Negatives – being denied opportunities they rightfully deserve.
- So, even though accuracy looks high (e.g., 92.1% for group 0 in the table), the model's errors can still create bias and unfair advantages.

Equal Opportunity (True Positive Rate – TPR)

- TPR tells us how good the model is at recognising people who truly deserve a positive outcome.
- It's also called Recall: the percentage of actual positives correctly predicted as positive.

For example:

- **Loans:** % of good customers who actually got the loan.
- **Hiring:** % of qualified candidates who were offered jobs.

$$TPR = \frac{TP}{TP + FN}$$

- **The denominator** = everyone who deserved the opportunity.
- **The numerator** = those who deserved and received it.

Fairness goal: *Equal TPR across all groups → everyone has an equal opportunity to benefit.*

Equal Opportunity (True Positive Rate – TPR)

	1	0	Ratio
Race	61.1%	53.3%	0.87
Sex	63.2%	44.3%	0.70

- **1** = Privileged group **0** = Unprivileged group
- The unprivileged group is more likely to face unfair or biased decisions from the model.
- TPR (True Positive Rate) measures how many people who *should benefit* actually *did benefit*.
- In this example, TPR values are lower for the unprivileged group, suggesting fewer members of this group are being correctly recognised or rewarded by the model.

For instance:

- ***Race: 61.1% (privileged) vs 53.3% (unprivileged)***
- ***Sex: 63.2% (privileged) vs 44.3% (unprivileged)***
- This means a smaller percentage of high-income earners in the unprivileged group were correctly predicted as having high income.
- The ratio shows how the unprivileged group's TPR compares to the privileged group.

Ideally, it should be close to 1 for fairness.

False Negative Rate (FNR)

	1	0	Ratio
Race	38.9%	46.7%	1.20
Sex	36.8%	55.7%	1.52

- FNR shows how many real positives your model missed.
- It's the percentage of people who should have been predicted positive but weren't.
- ***Example:*** *People who actually qualified for a loan but were rejected*
- ***A high FNR for the unprivileged group means the model misses more opportunities for them — that's unfair!***
- For Sex, FNR = 55.7% → means more women were wrongly denied the positive outcome (e.g., a loan or job).
- For Race, FNR = 46.7% → still unfair, but slightly better.
- The ratio (1.52) shows the model is 52% more likely to miss positives for one group compared to another.

Equalised Odds (False Positive Rate – FPR)

- Equalised odds means fairness requires both:
- Equal **True Positive Rates (TPR)** → equal chances to *rightfully benefit*, **and**
- Equal **False Positive Rates (FPR)** → equal chances to *wrongfully benefit*.

	1	0	Ratio
Race	8.1%	3.9%	0.48
Sex	10.9%	1.7%	0.16

From the figure:

- **Race:** 8.1% vs 3.9% → the privileged group benefited almost twice as often by mistake.
- **Sex:** 10.9% vs 1.7% → the privileged group benefited much more unfairly.

A truly fair model ensures that both groups have similar opportunities — whether those outcomes are right or wrong.

Disparate Impact (DI)

- DI checks if both groups get similar chances to be picked by the model.
- It doesn't matter if the prediction is right or wrong – just who gets the "Yes".

	1	0	Ratio
Race	22.0%	11.7%	0.53
Sex	27.3%	6.6%	0.24

$$\%\ \text{predicted as positive (PPP)} = \frac{TP + FP}{N}$$

This tells us the percentage of people who got a positive prediction (for example, loan approved, job offer, university admission).

If the ratio is below 0.8, the model is unfair to the unprivileged group. **This is called the "80% rule."**

Example (from figure):

- **Race: 11.7% vs 22.0% → 0.53 → unfair**
- **Sex: 6.6% vs 27.3% → 0.24 → unfair**

A model deciding who gets scholarships.
If 20 out of 100 privileged students get one but only 10 out of 100 unprivileged students do — the unprivileged group is missing out, even if both groups had good grades.

Questions

